Spatial Databases GROUP PROJECT REPORT (LAQP for spatial data)

Name: 汪詩揚, 謝豐廷, 鄭傳脩

Sid: 114598020, 114598036, 113598022

Date:12/30/2025

National Taipei University of Technology
Computer Science and Information Engineering

OUTLINE

- Introduction
- Literature review and related works
- Problem statement
- Proposed models (approaches)
- Experiments
- Conclusions
- Others

INTRODUCTION

- **Big Data Challenges:** Traditional queries are too slow on millions of rows of data.
- **Solution:** Use **AQP (Approximate Query Processing)** to significantly reduce query time within an acceptable error range.
- We use LAQP (Learning-based AQP), which combined **sampling-based, pre-computed aggregations and ML**, to achieve low error with small samples on spatial data.

INTRODUCTION

- **Experiment** on **POWER** dataset (as in paper) and extended to spatial data using **Uber Pickups NYC** dataset for queries on time + lat/lon ranges.
- Compare the performance of **AQP++** (Sampling & Pre-computation) , **SAQP** (Sampling-based AQP) and **LAQP** (Learning-based AQP) on the **SUM** and **COUNT** aggregation function.
- **Streamlit demo** for interactive bbox/time queries.
- Our code is on github.

LITERATURE REVIEW AND RELATED WORKS

- **AQP++** is an interactive analytics framework that bridges sampling-based approximate query processing (AQP) and aggregate precomputation. Instead of directly estimating query results from samples, AQP++ estimates the difference between a user query and a precomputed aggregate, and then corrects the result using the exact aggregate value. By exploiting the correlation between queries and precomputed aggregates, AQP++ achieves higher accuracy than traditional AQP with modest preprocessing cost.[1]

- LAQP is a learning-based approximate query processing framework that combines sampling-based AQP, precomputed aggregations, and machine learning. It learns a regression model from historical queries to predict the sampling error of a new query and selects an error-similar precomputed query instead of a range-similar one. By estimating query results using this error correction strategy, LAQP achieves higher accuracy with a small sample size while still providing statistical error guarantees.[2]

PROBLEM STATEMENT

- Existing methods (SAQP, AQP++, ML) have shortcomings like no error guarantees or limited query support.

- Spatial data like Uber Pickups are multi-dimensional ranges, sampling alone is inaccurate on skewed locations, using hybrid AQP for better performance.

PROPOSED MODELS (APPROACHES)

- AQP++: The AQP++ estimates the new query based on its 'range-similar' pre-computed query, whose predicate range is the most similar to the given query.

- LAQP: Unlike AQP++, which selects the baseline query based on predicate range similarity, LAQP chooses a pre-computed query whose sampling-based estimation error is most similar to the predicted error of the new query. This error-driven selection is enabled by a learned regression model trained on the query log, making LAQP more robust for high-dimensional and skewed data.

PROPOSED MODELS (APPROACHES)

- AQP++ Concepts

$$q(\mathcal{D}) \approx pre(\mathcal{D}) + \left(\hat{q}(\mathcal{S}) - \hat{pre}(\mathcal{S})\right)$$

- LAQP Concepts

$$est(q) = R(q_{opt}) + EST(q) - EST(q_{opt}),$$

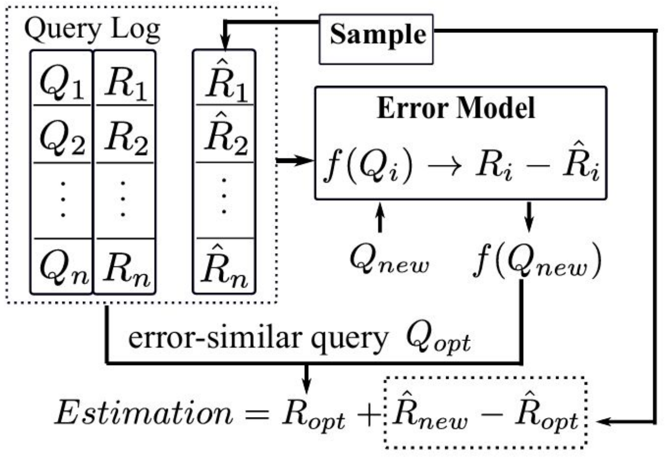
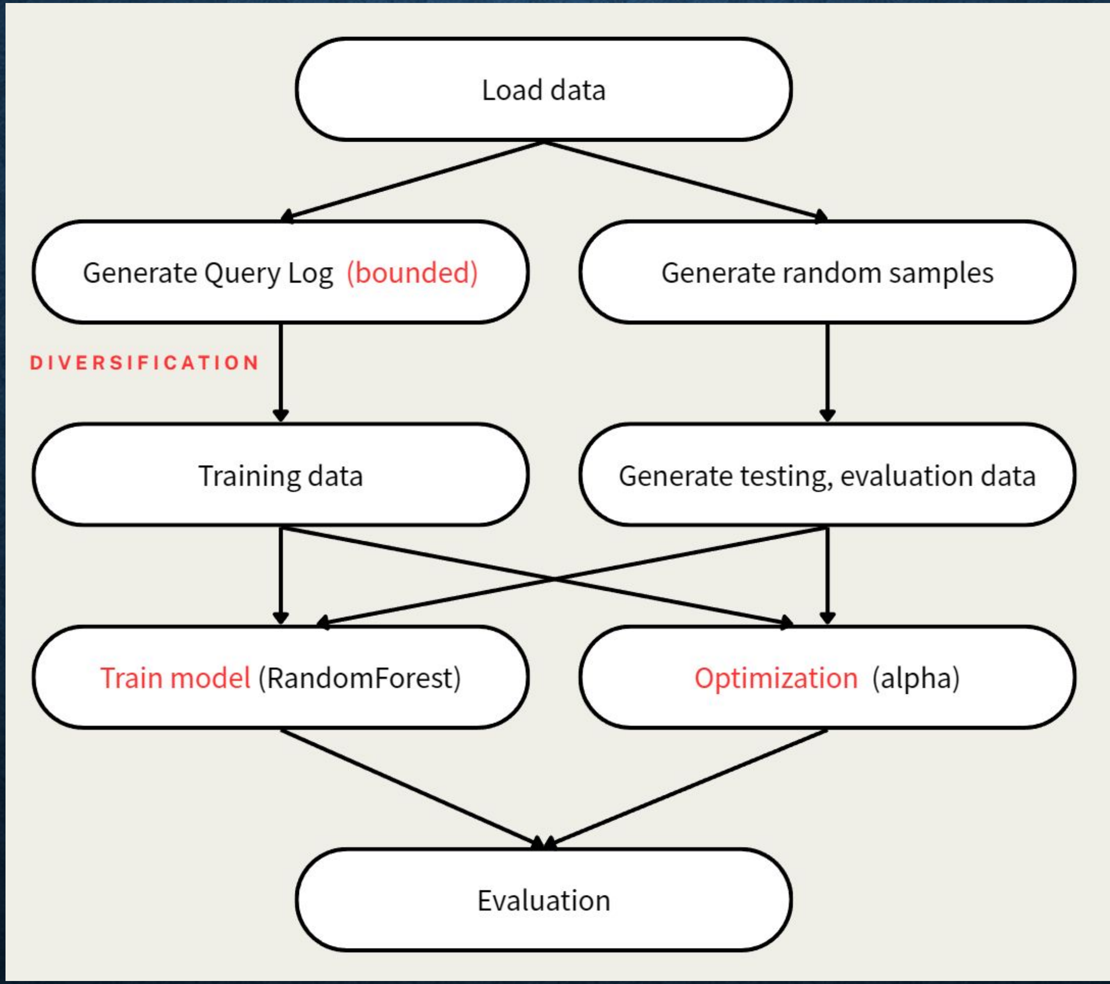


Figure 1: The framework of LAQP

EXPERIMENTS (Architecture)



EXPERIMENTS (Data Preprocessing)

```
16/12/2006;17:24:00;4.216;0.418;234.840;18.400;0.000;1.000;17.000
16/12/2006;17:25:00;5.360;0.436;233.630;23.000;0.000;1.000;16.000
16/12/2006;17:26:00;5.374;0.498;233.290;23.000;0.000;2.000;17.000
16/12/2006;17:27:00;5.388;0.502;233.740;23.000;0.000;1.000;17.000
16/12/2006;17:28:00;3.666;0.528;235.680;15.800;0.000;1.000;17.000
```

```
Dataset loaded: 2049280 rows
First 5 data:     timestamp  Global_active_power  Global_reactive_power  Voltage  \
0        0.0                4.216                  0.418   234.84
1       60.0                5.360                  0.436   233.63
2      120.0                5.374                  0.498   233.29
3      180.0                5.388                  0.502   233.74
4      240.0                3.666                  0.528   235.68

   Global_intensity  Sub_metering_1  Sub_metering_2  Sub_metering_3
0              18.4             0.0             1.0            17.0
1              23.0             0.0             1.0            16.0
2              23.0             0.0             2.0            17.0
3              23.0             0.0             1.0            17.0
4              15.8             0.0             1.0            17.0
```

POWER

```
4/1/2014 0:11:00,40.769,-73.9549,
4/1/2014 0:17:00,40.7267,-74.0345,
4/1/2014 0:21:00,40.7316,-73.9873,
4/1/2014 0:28:00,40.7588,-73.9776,
4/1/2014 0:33:00,40.7594,-73.9722,
```

```
Dataset loaded: 4534327 rows
   timestamp      Lat      Lon
0      660.0  40.7690 -73.9549
1     1020.0  40.7267 -74.0345
2     1260.0  40.7316 -73.9873
3     1680.0  40.7588 -73.9776
4     1980.0  40.7594 -73.9722
```

Uber

EXPERIMENTS (Diversification)

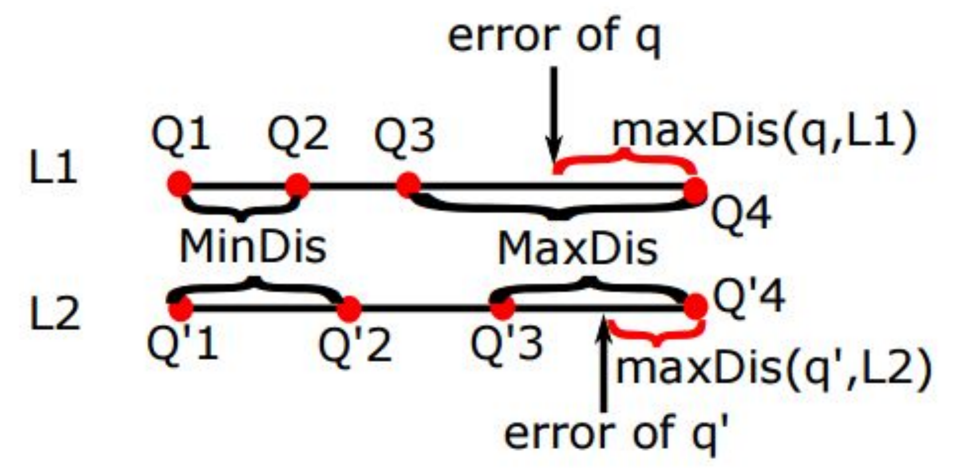


Figure 2: Diversification

EXPERIMENTS (Bounded)

“In order to avoid most of the multi-dimensional query results to be zero, we limit the range to generate the boundaries in each dimension. The left boundary of the range in each dimension is randomly chosen from the first quarter of the entire value range of the corresponding attribute. Similarly, each right boundary is randomly chosen from the last quarter of the corresponding attribute range.”

EXPERIMENTS (Optimization)

$$opt(q, \alpha, \beta) = \underset{Q_i \in Train}{\arg\min}\ \alpha \cdot EDis(q, Q_i) + \beta \cdot RDis(q, Q_i)$$

$$EDis(Q_i, Q_j) = |error_{Q_i} - error_{Q_j}|^2$$

$$RDis(Q_i, Q_j) = \frac{\sum_{x \in d}^{(l_{i,x} - l_{j,x})^2 + (r_{i,x} - r_{j,x})^2}}{2d}$$

$$s.t.\ \alpha + \beta = 1, 0 \le \alpha \le 1, 0 \le \beta \le 1.$$

$$Dis(q, Q_i) = \alpha \cdot EDis(q, Q_i) + \beta \cdot RDis(q, Q_i)$$

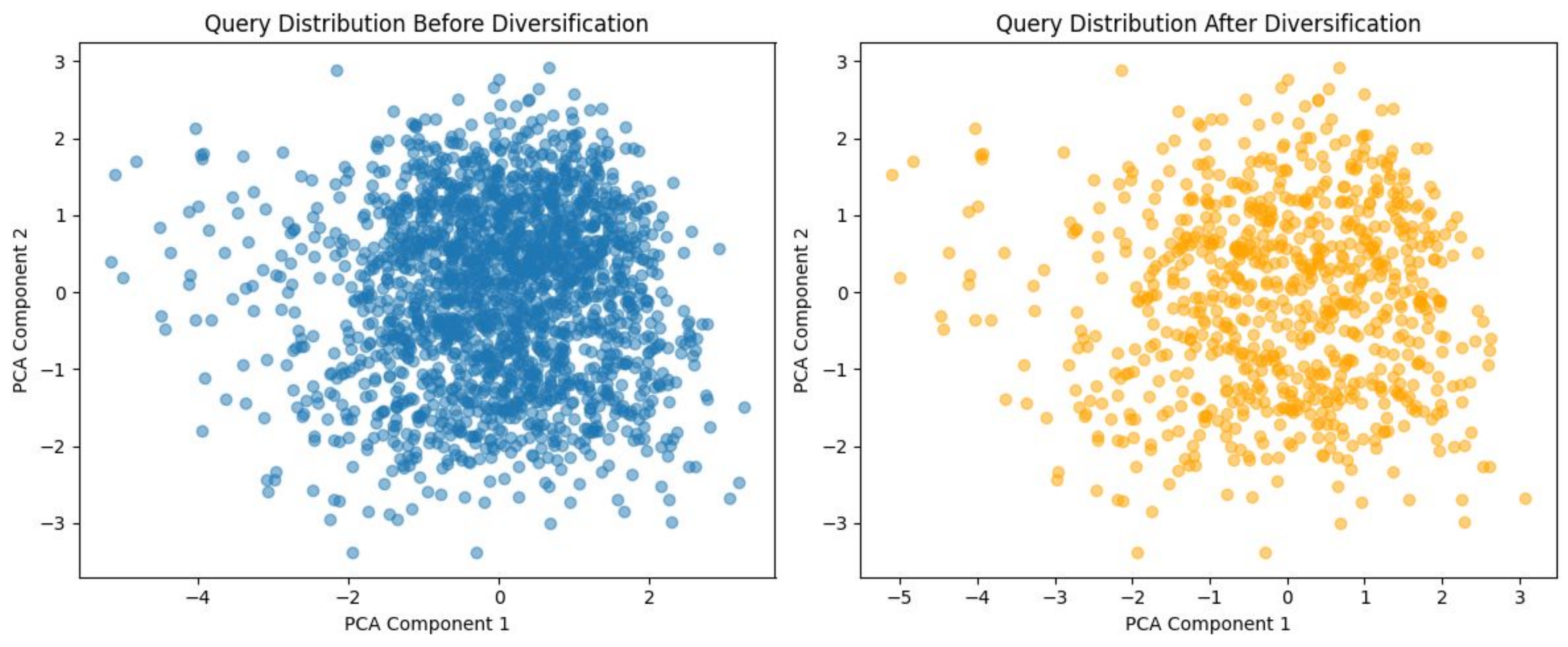
```
Optimized alpha: 0.987
```

EXPERIMENTS (POWER)

```
{'query': {'timestamp': (24775624.50534664, 109077562.53846367),
  'Global_reactive_power': (0.21508427287010004, 1.243410333793932),
  'Voltage': (225.114600554981, 249.6732681343024),
  'Global_intensity': (3.1425672365870523, 41.70827352541356),
  'Sub_metering_1': (6.906678248160892, 84.90158493861541),
  'Sub_metering_2': (1.877898024454443, 67.46303711982367),
  'Sub_metering_3': (4.241749905529564, 28.020941984101842)},
 'exact': 13715.400000000001,
 'estimate': 9301.68192,
 'error': 4413.718080000001}
```

Sample	Training data	Testing data	max_depth	Evaluation data	Alpha
2000	800 / 2000	100	3	100	1.0

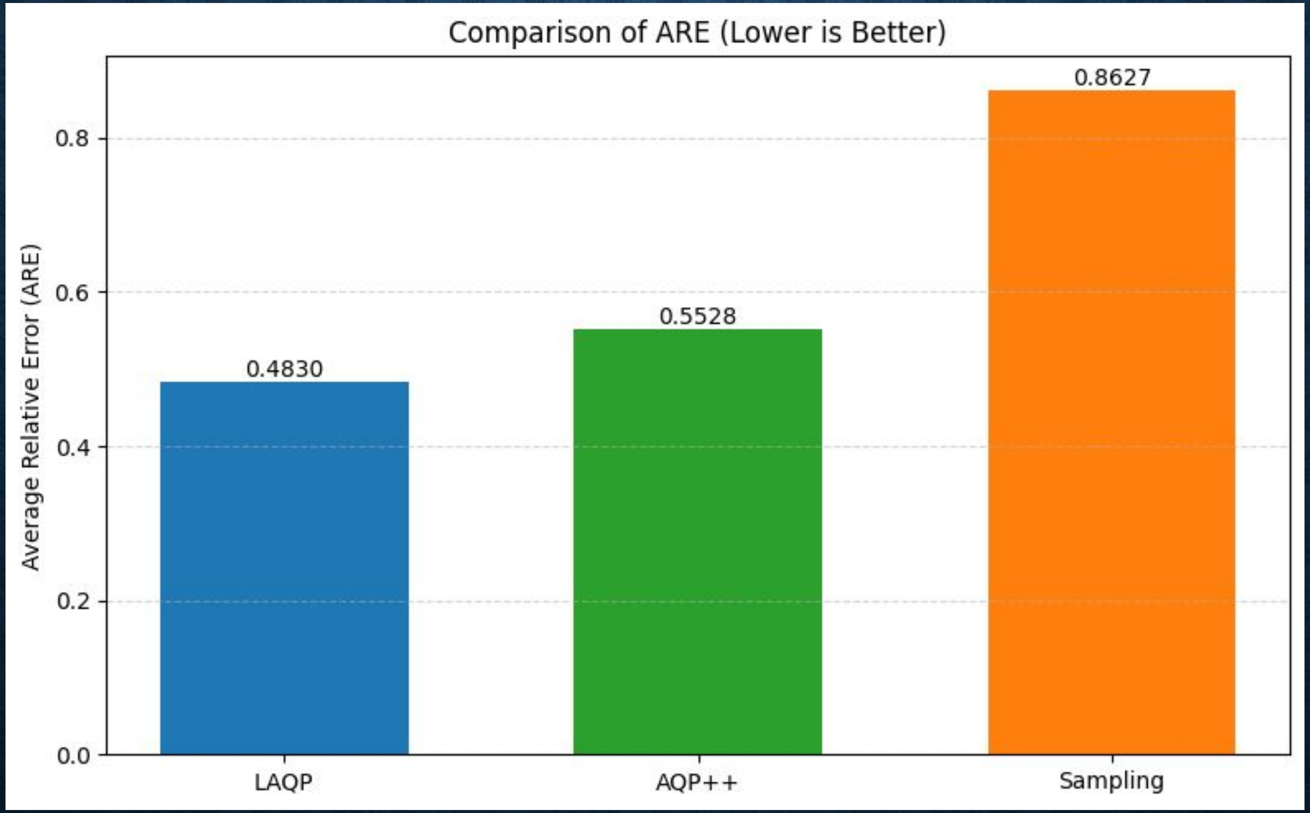
EXPERIMENTS (Diversification)




```
Diversified query log: selected 800 / 2000 entries
```

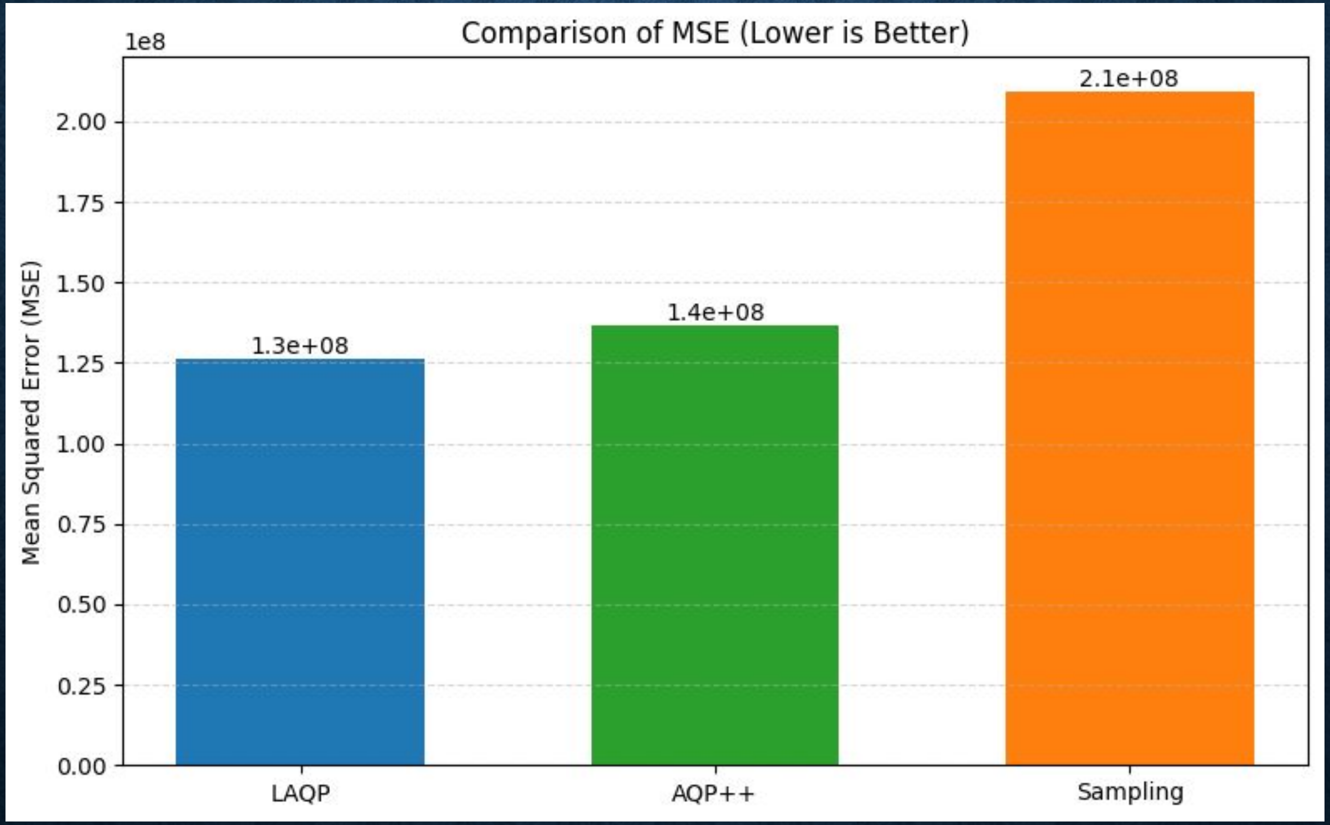
EXPERIMENTS (Results)

power dataset (ARE)



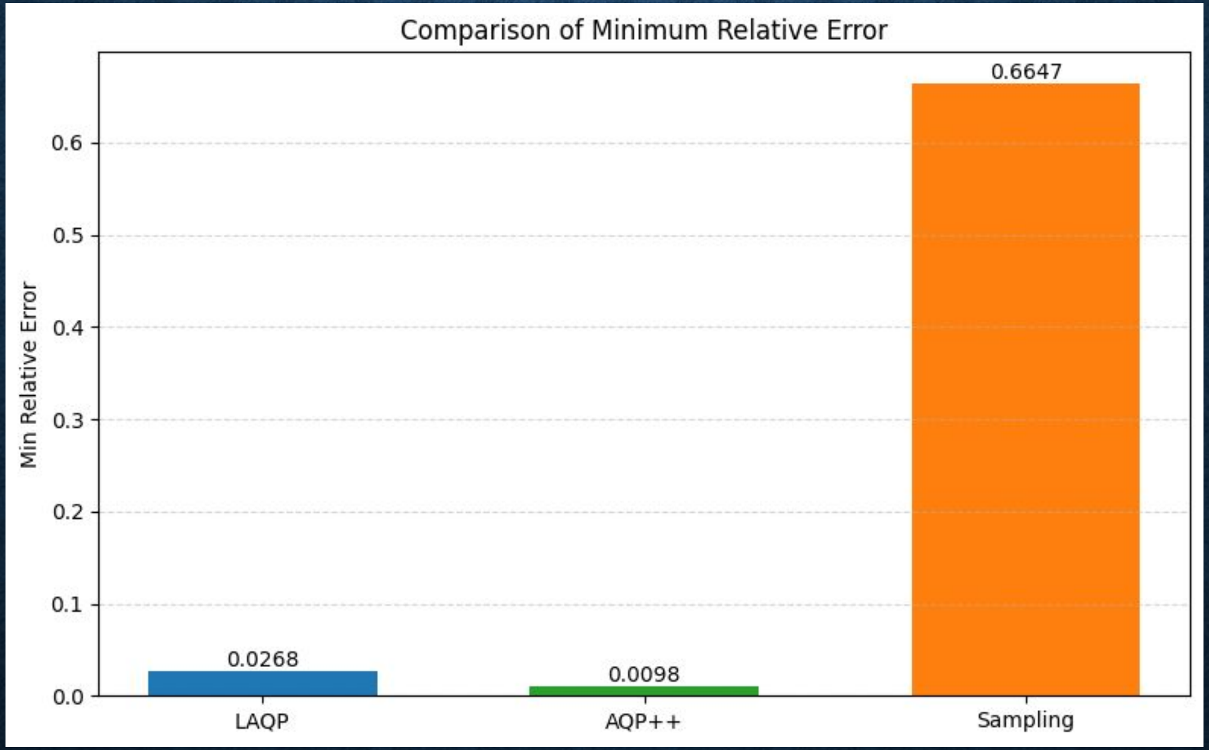
EXPERIMENTS (Results)

power dataset (MSE)



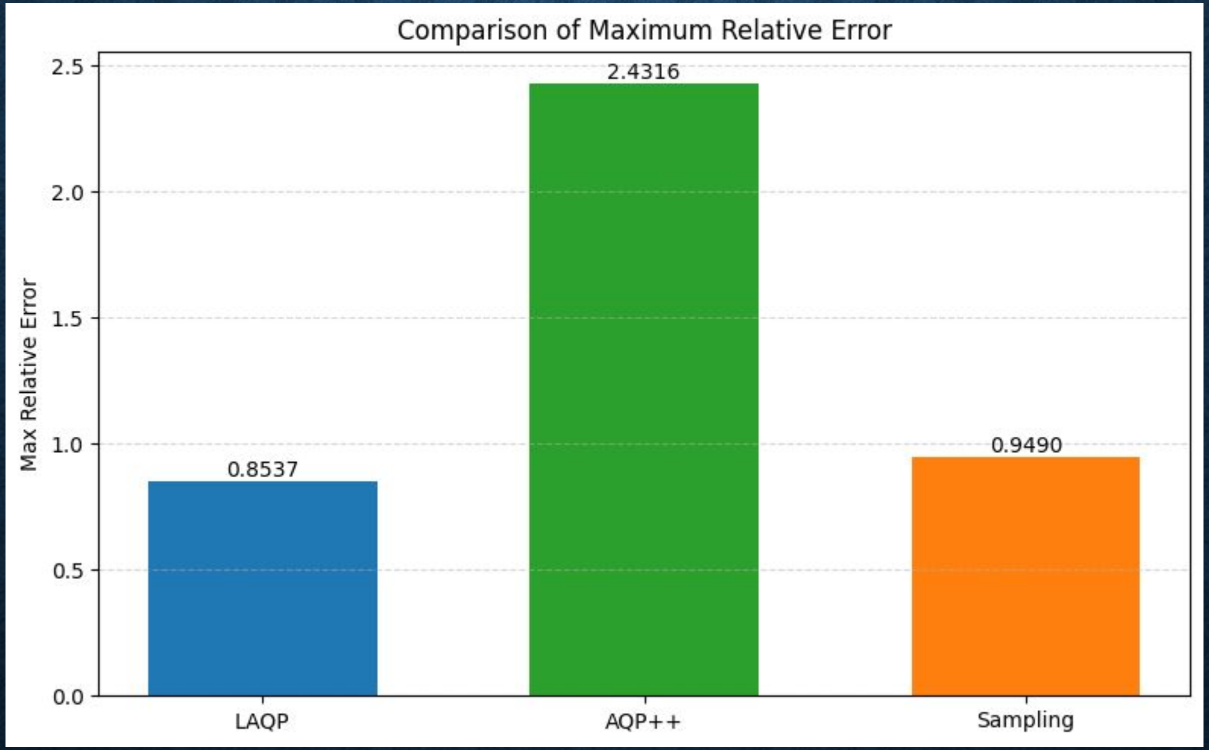
EXPERIMENTS (Results)

power dataset(min relative error)



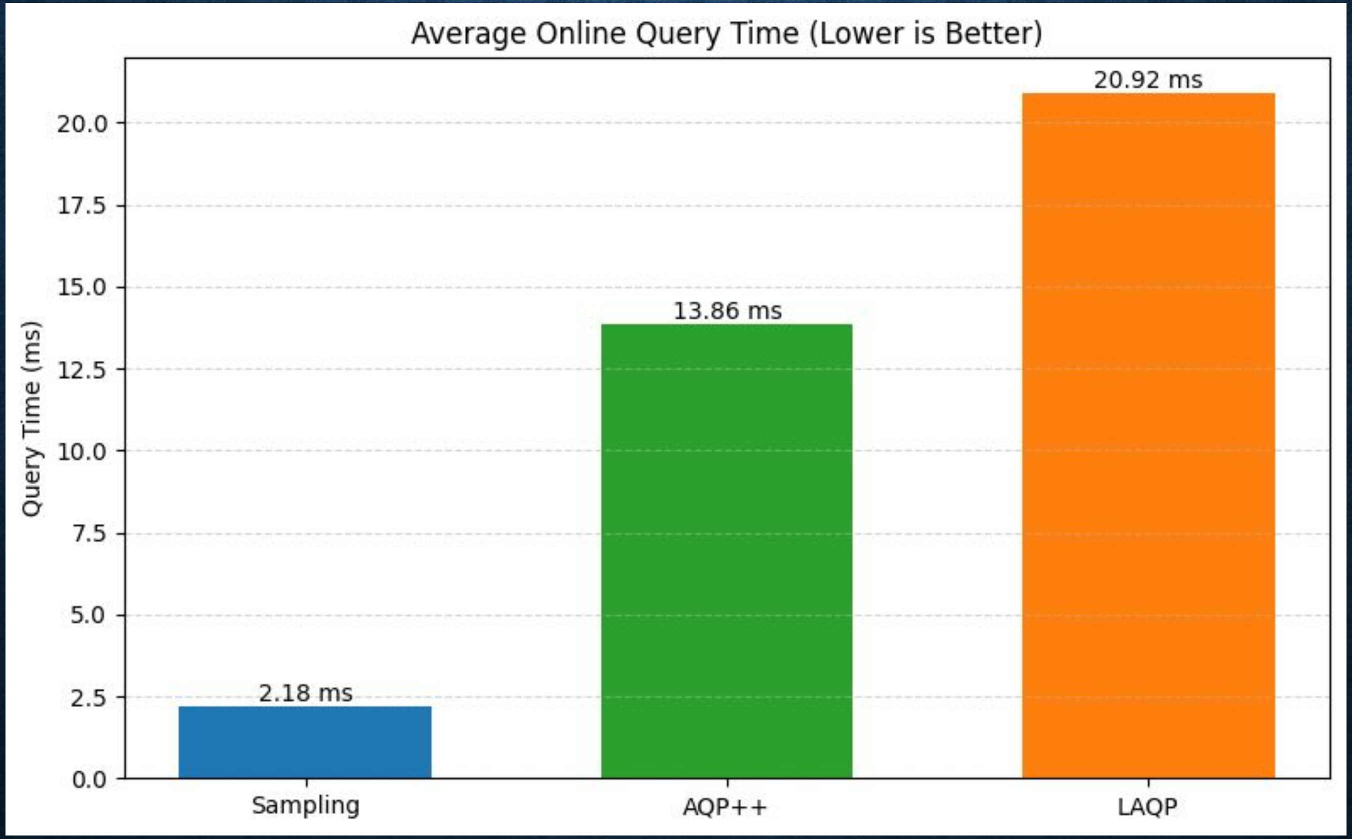
EXPERIMENTS (Results)

power dataset(max relative error)



EXPERIMENTS (Results)

Time

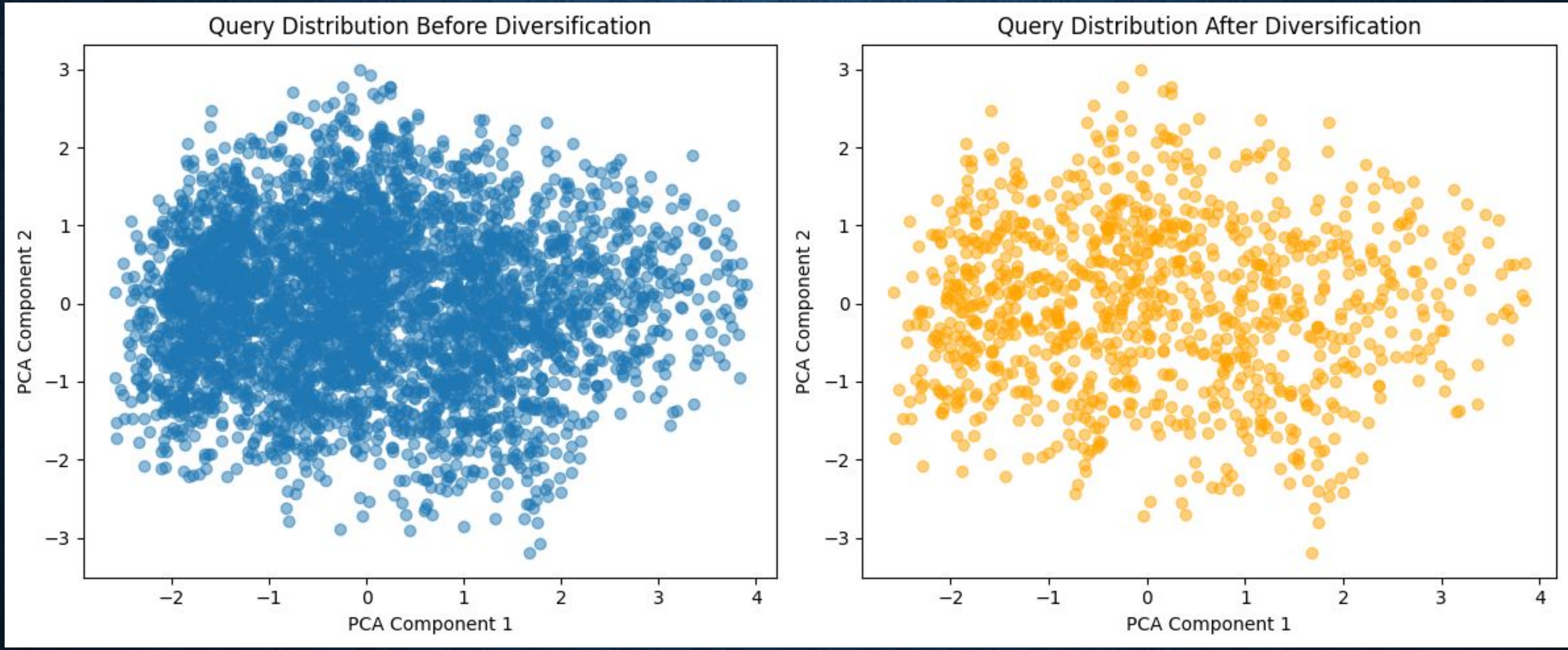


EXPERIMENTS (Uber)


```
{'query': {'timestamp': (532457.7783156144, 15650249.140553676),
  'Lat': (40.01059159082038, 41.90556339339713),
  'Lon': (-74.26716490851912, -72.61707661728755)},
 'exact': 4354927,
 'estimate': 4360525.9650883265,
 'error': -5598.965088326484}
```


Sample	Training data	Testing data	max_depth	Evaluation data	Alpha
45343	1000 / 4000	250	10	250[:100]	1.0

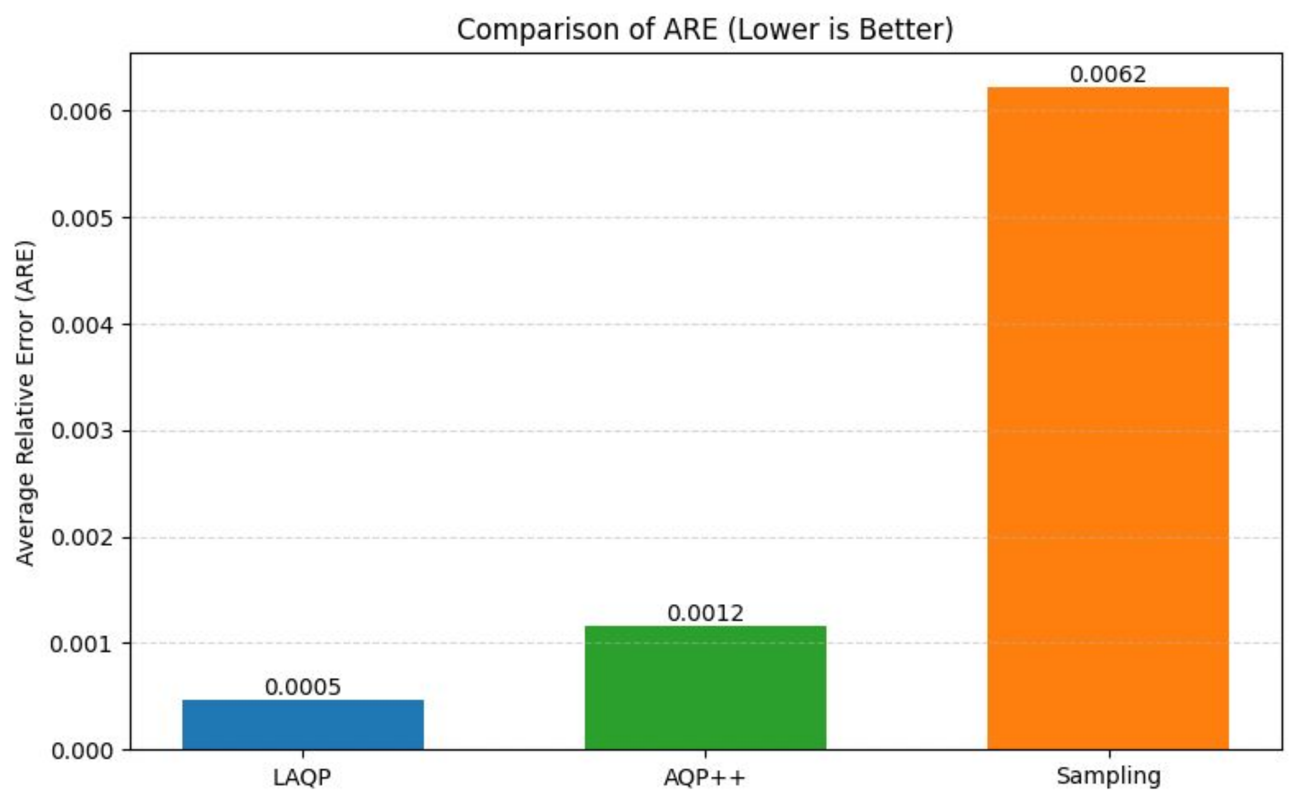
EXPERIMENTS (Diversification)



`Diversified query log: selected 1000 / 4000 entries`

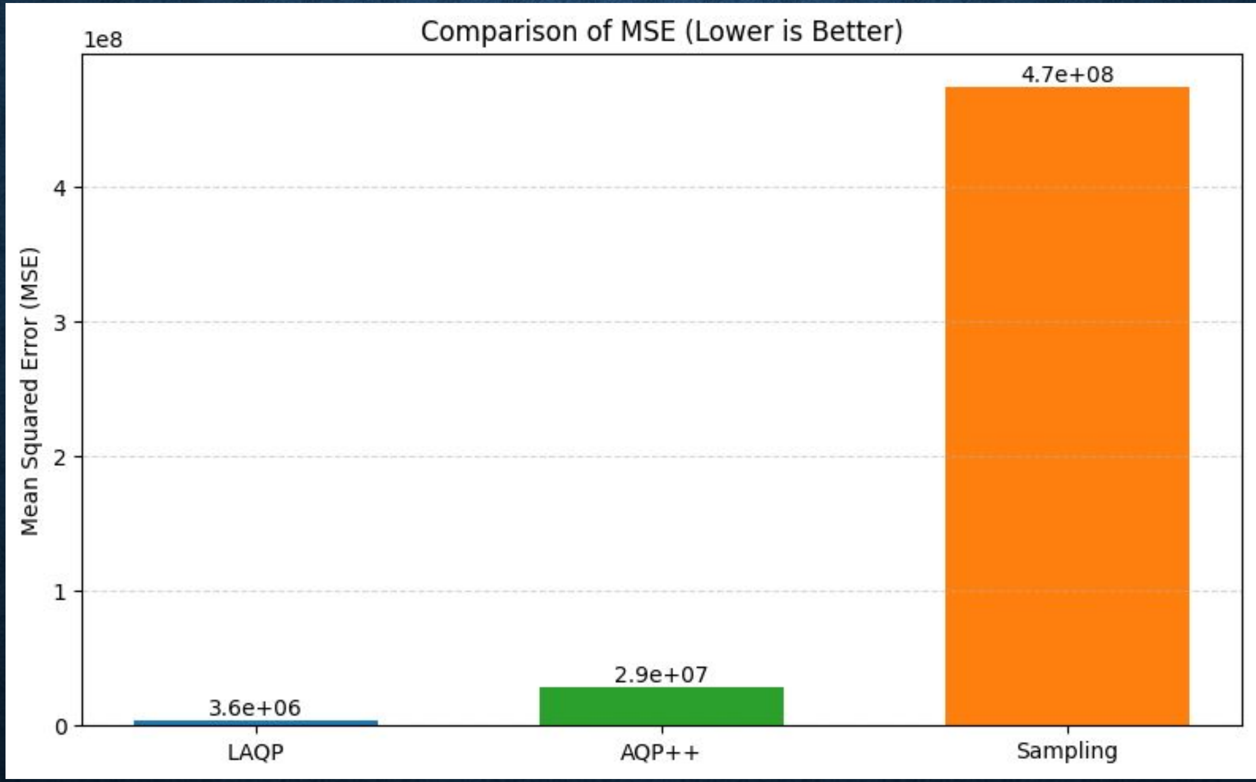
EXPERIMENTS (Results)

uber dataset (ARE)



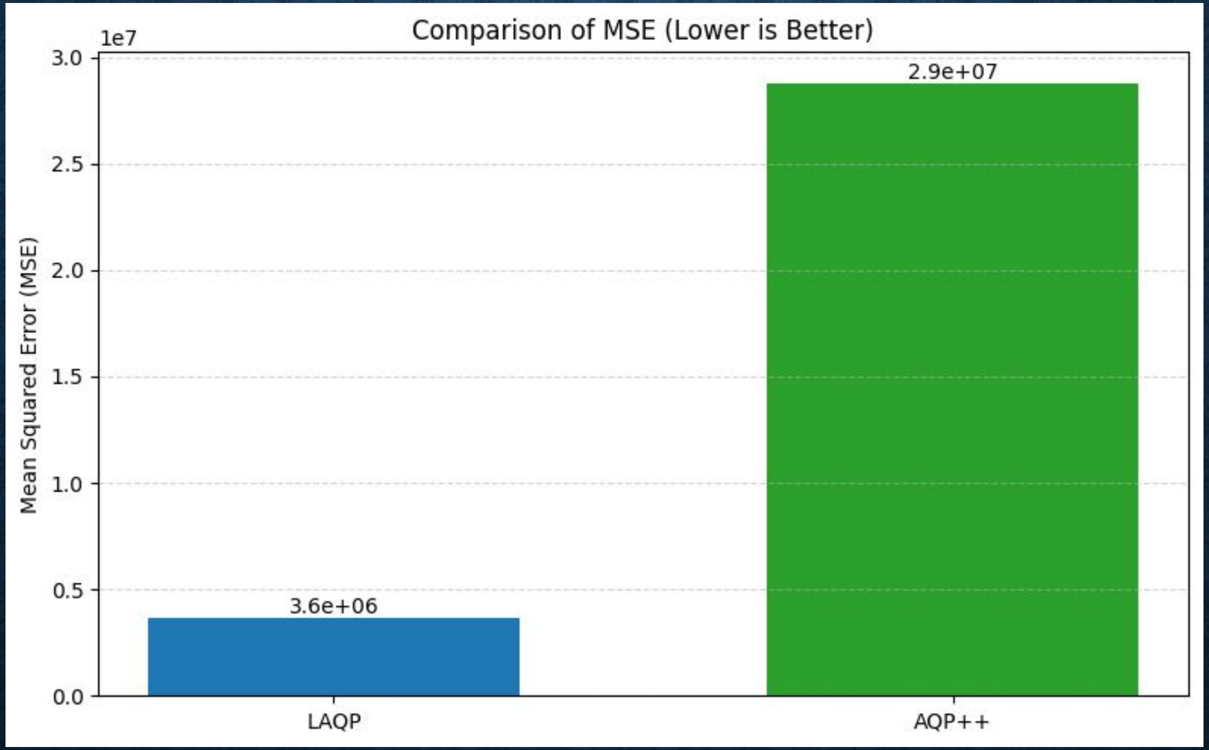
EXPERIMENTS (Results)

uber dataset(MSE)



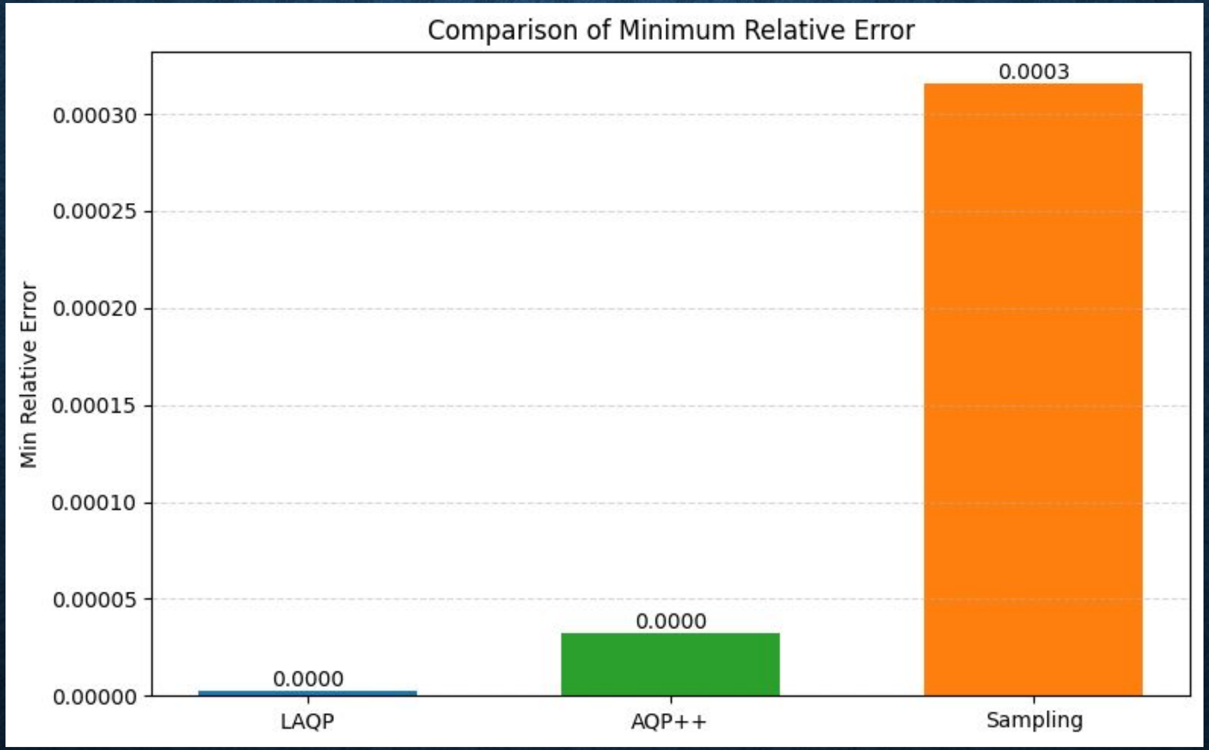
EXPERIMENTS (Results)

uber dataset(MSE)



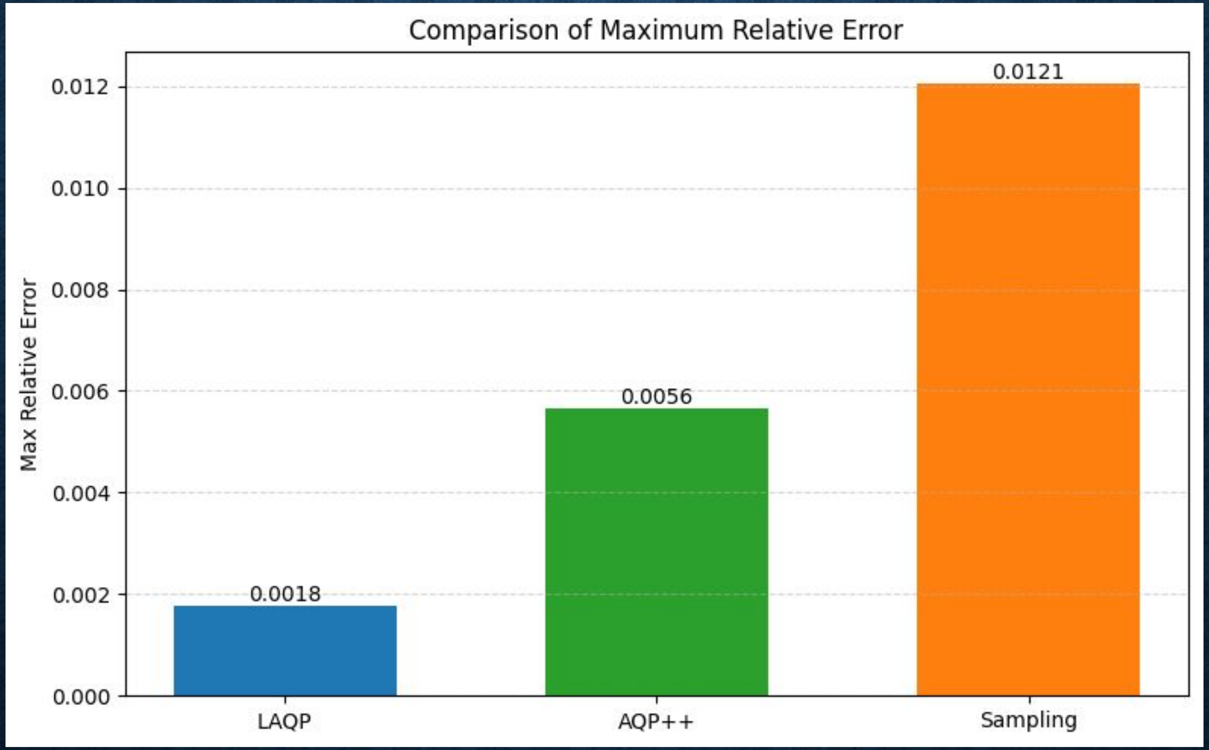
EXPERIMENTS (Results)

uber dataset(min relative error)



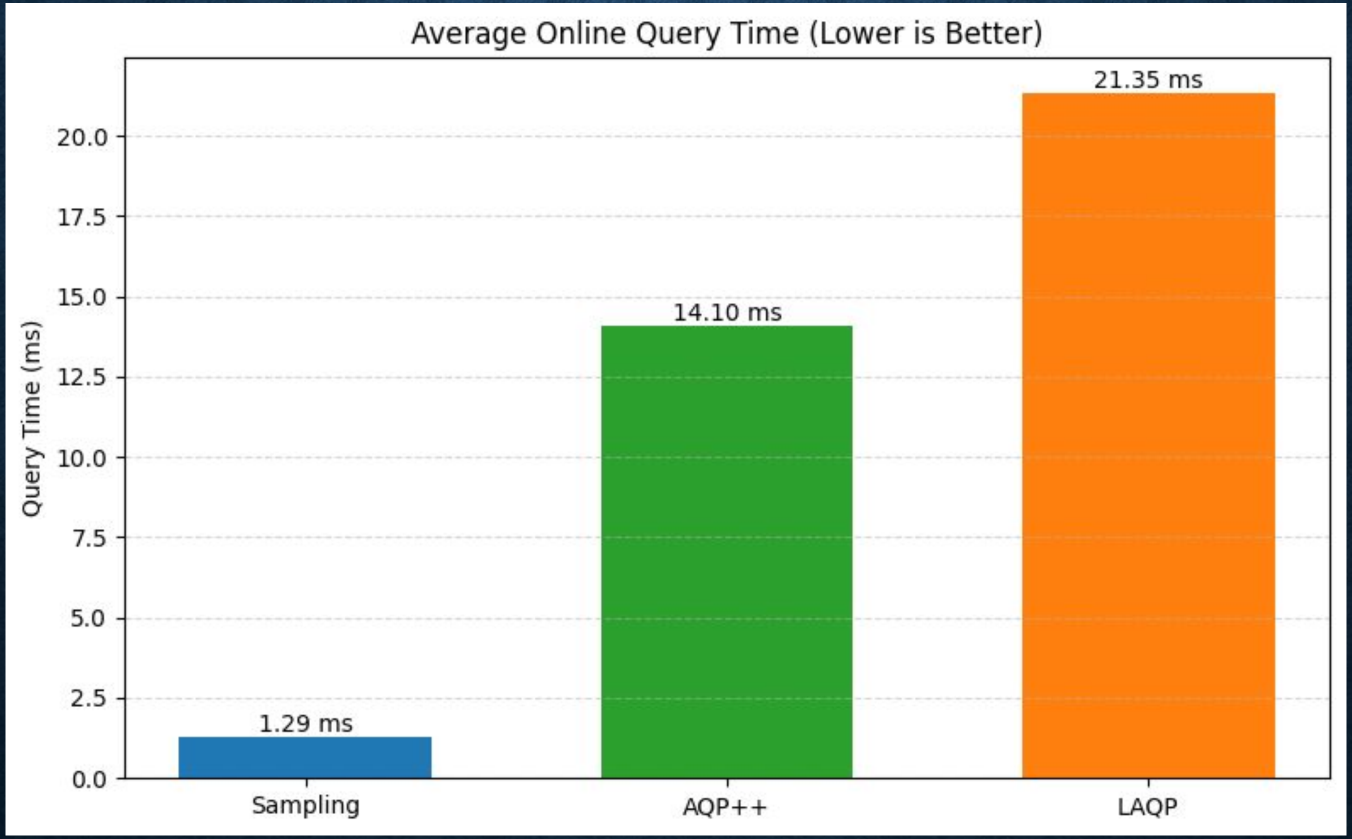
EXPERIMENTS (Results)

uber dataset(max relative error)



EXPERIMENTS (Results)

Time



CONCLUSIONS

- Successfully reproduced LAQP, extended to spatial COUNT on Uber Pickups, built interactive demo proving practical value.
- Experiment shows LAQP has better performance than AQP++ and SAQP.
- Future work: Support more aggs (avg) on datasets. Compare with more methods like DBest. Add more performance comparison like cost.

OTHERS

	week9	week10	week11	week12	week13	week14	week15	week16	week17	Done
Environment setup										V
Reproduce LAQP										V
Spatial adaptation + Training model										V
Interactive System										V
Experimental evaluation + Demo										V

OTHERS

Name / Student ID	role	workload
汪詩揚 / 114598020	LAQP, Demo Web, github, PTT	55%
謝豐廷 / 114598036	AQP++, github, PTT	30%
鄭傳脩 / 113598022	github, PTT	15%

References

[1] Peng, J., Zhang, D., Wang, J., Pei, J., 2018. AQP++: connecting approximate query processing with aggregate precomputation for interactive analytics, in: Proceedings of the 2018 International Conference on Management of Data, SIGMOD Conference 2018, Houston, TX, USA, June 10-15, 2018, pp. 1477–1492. URL: https://doi.org/10.1145/3183713.3183747, doi:10.1145/3183713.3183747.

[2] Zhang, Meifan, and Hongzhi Wang. "LAQP: Learning-based approximate query processing." Information Sciences 546 (2021): 1113-1134.

DEMO

Demo builds a streamlit website on localhost.

User can choose city like Brooklyn, Manhattan and strict the latitude and longitude.

There is a calendar which bounds Date/Time according to the Uber Pickups dataset.

Also, user can draw a bbox to customize query.

140.124.182.132:8501 (Only when code is running, user can run their own demo by following github)

DEMO

LAQP Uber Pickups COUNT Demo

Select City Region

Manhattan

Apply City Region

Start Date

2014/04/01

End Date

2014/09/30

Lat Range (BBox)

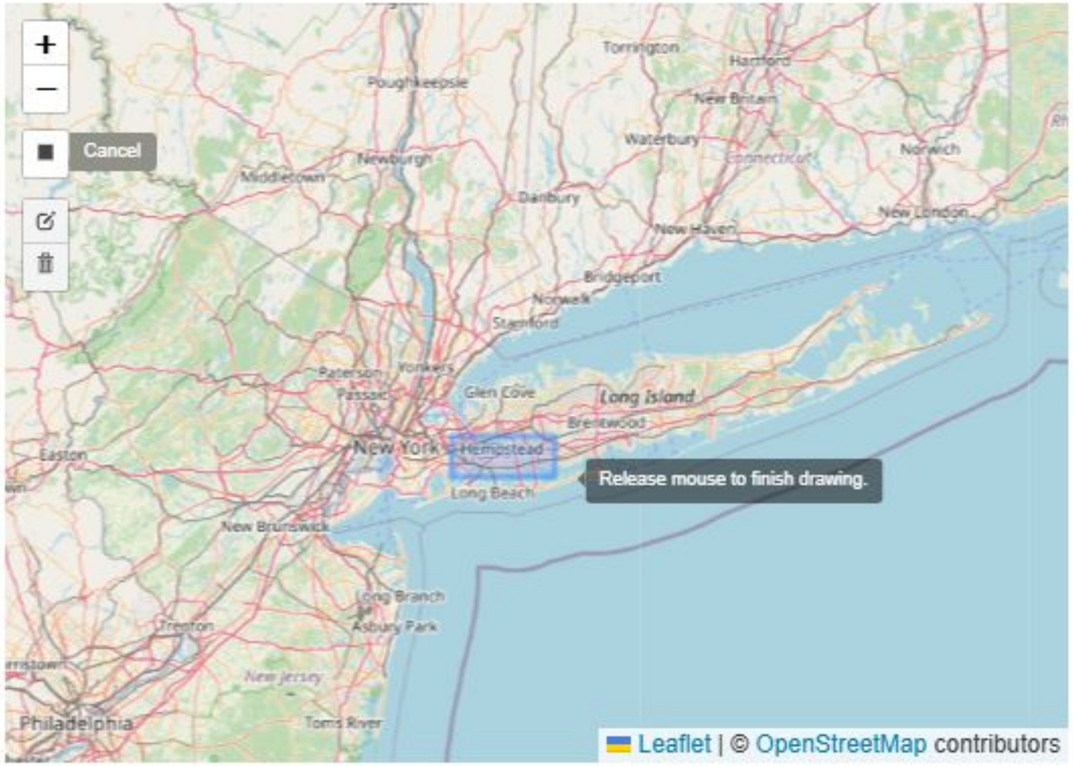
40.70 40.82

Lon Range (BBox)

-74.02 -73.93

Generate Random Lat, Lon

Draw Custom BBox on Map



Query BBox Preview (Matching Points Up to 1000)

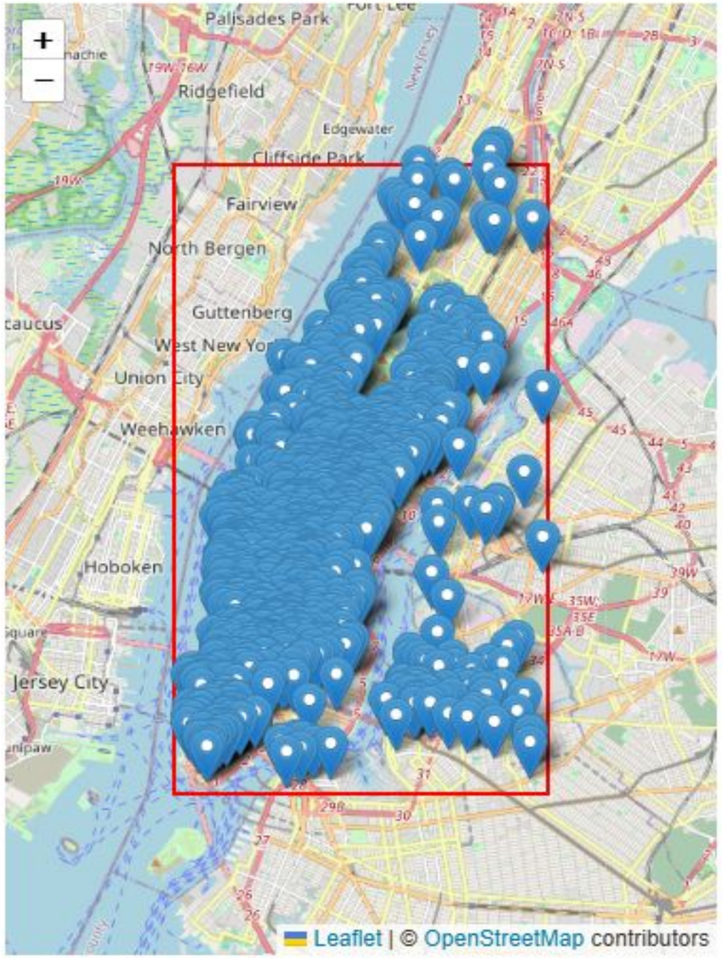
Query Search

Show Matching Points on Map

Exact COUNT: 3662230

LAQP Approx COUNT: 3661009

Difference: 1220.9859 (Rel Error: 0.0003)



Thank you